

AI-Powered Travel Booking and Planning Website

Software Project Proposal

*with the supervision of Teacher Nguyễn Huy Khánh, Teacher
Trần Duy Quang, Teacher Trần Văn Quý*

22127410 - Lưu Thanh Thúy

22127435 - Võ Lê Việt Tú

22127444 - Trần Thị Cát Tường

University of Science
Faculty of Information Technology
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Mục lục

1	Project Overview	2
2	Executive Summary	2
3	Introduction	3
3.1	Project Background	3
3.2	Problem Statement	3
3.3	Project Objectives	3
3.4	Market Analysis and Competitive Advantage	4
3.4.1	Competitor Review	4
3.4.2	Unique Value Proposition	4
4	Core Features	5
4.1	User-Facing Features	5
4.2	Backend & Technical Features	5
5	Technical Challenges and Learning Goals	6
5.1	Technical Challenges	6
5.2	Learning Goals	6
5.3	Feasibility and Timeline	7
6	Expected Outcome	7
7	Future Improvements	7
8	Evaluation Criteria Alignment	8
9	Conclusion	8

1 Project Overview

Item	Information
Product Title	AI-Powered Travel Booking and Planning Website
Project Manager	Lưu Thanh Thuý
Project Duration	10 weeks

2 Executive Summary

This proposal outlines the development of an **AI-powered travel platform** designed to simplify the process of searching, planning, and booking travel services — including hotels, flights, and tours — through a **natural conversational interface**.

Currently, users must rely on multiple platforms to organize their trips, making the process time-consuming, fragmented, and impersonal. The proposed system introduces an **AI Agent** that allows users to describe their travel needs in natural language and receive **customized itineraries** instantly. It will integrate **real-time booking APIs** to retrieve live data and enable basic reservation functionality within the same platform.

The system will generate **personalized 3–7 day travel plans** based on user preferences, budget, and travel history, while continuously improving recommendations over time through user profiling and feedback learning.

Expected benefits include: - Reducing the time and complexity involved in trip planning. - Increasing the relevance and personalization of travel recommendations. - Improving booking conversion rates through an intelligent, conversational experience.

Beyond functionality, this project demonstrates the **practical application of Large Language Models (LLMs)** in a real-world tourism product, paving the way for future AI-enhanced features.

The final deliverable will be a **web-based prototype** developed within **6–8 weeks**, featuring the AI chat module, core search and booking functions, and an itinerary-generation engine — ready for testing, validation, and further expansion.

3 Introduction

3.1 Project Background

The modern travel industry has evolved rapidly with digitalization, offering users countless online tools for booking hotels, flights, and activities. However, despite this abundance of options, **trip planning remains a fragmented and time-consuming process**. Travelers typically switch between multiple platforms — such as Agoda, Booking.com, Expedia, and Skyscanner — to compare prices, read reviews, and manually organize their itineraries.

At the same time, recent advancements in **Artificial Intelligence (AI)**, particularly **Large Language Models (LLMs)**, have introduced new opportunities for intelligent automation and personalization. These models can understand user intent, context, and preferences through natural conversation, enabling a more intuitive and interactive approach to problem-solving.

This project is motivated by the opportunity to combine **AI conversational capabilities** with **real-world booking functionalities** to create a **seamless, all-in-one travel experience**. By integrating AI-driven trip planning with booking APIs, the platform aims to simplify how users plan their vacations — making the process faster, smarter, and more personalized.

3.2 Problem Statement

Currently, travelers face several challenges when organizing a trip:

- Searching for hotels, flights, and local attractions across multiple websites.
- Building an itinerary that fits time, budget, and preferences.
- Comparing ratings, prices, and reviews manually.

This fragmented workflow makes trip planning **complex, time-consuming, and impersonal**.

Existing booking platforms such as **Agoda** and **Booking.com** provide strong search and reservation systems but **lack conversational interaction** and **personalized itinerary generation**. They do not allow users to simply “talk” to the system about their travel needs or receive a tailored plan automatically.

For example, if a user asks: > “I have five days in Da Lat with a \$400 budget. Can you suggest an itinerary and a place to stay?”

Traditional booking systems can only display hotel lists; they cannot understand context, optimize travel schedules, or build adaptive itineraries. This highlights a significant gap where **AI-powered natural language systems** can bring meaningful improvement.

3.3 Project Objectives

The goal of this project is to build an **AI-integrated travel booking website** that provides users with a **personalized and interactive trip planning experience**. Specifically, the system aims to:

- Develop a **web-based platform** where users can **converse directly with an AI Agent** to describe travel goals and preferences.
- Automatically **generate personalized itineraries** — including day-by-day schedules, accommodation options, and recommended activities — based on user interests, time, and budget.
- Integrate **real-time booking APIs** to fetch updated hotel and flight data, allowing users to **search and make reservations directly within the chat interface**.
- Simplify the overall trip planning process, reducing user effort while providing more relevant and contextual recommendations.

Ultimately, the platform seeks to replicate the experience of working with a **personal virtual travel consultant** that plans, recommends, and assists — all through natural conversation.

3.4 Market Analysis and Competitive Advantage

3.4.1 Competitor Review

Competitor	Key Features	Strengths	Weaknesses
Agoda	Hotel and flight booking, local deals	Intuitive interface, large data coverage, strong reputation	Lacks personalization and no AI-based planning or chat assistant
Booking.com	Accommodation booking, review system, filtering	Easy to use, global reach, reliable user base	No automated itinerary creation or conversational features
Trip Planner AI (Beta)	AI-based itinerary suggestions	Innovative idea leveraging AI planning	Not connected to real booking APIs, limited UX, lacks real-time data

3.4.2 Unique Value Proposition

Unlike existing platforms, this project's website is not merely a booking system — it is an **intelligent travel companion**.

It offers a new level of convenience and personalization through the following core advantages:

- **Context-Aware AI Assistance:** Understands user intent, preferences, and constraints (e.g., budget, travel dates, desired experiences) through natural conversation.
- **Personalized Itinerary Generation:** Automatically produces detailed, multi-day travel plans tailored to each user's unique profile and travel objectives.
- **Integrated Booking Experience:** Allows users to **search, plan, and book** within a single platform — eliminating the need to switch between multiple sites.

- **Conversational Interaction:** Users interact with the system naturally, asking questions and refining their plans in real time — just like speaking with a human travel consultant.

By combining **AI-driven planning** with **functional booking integration**, the proposed system fills a critical gap in today’s travel ecosystem. It transforms trip planning from a manual, tedious process into an **intuitive, conversational, and intelligent experience**.

4 Core Features

Internal Note (to be removed before final submission): Please update these sections with information relevant to your team’s specific scope and intended functionality.

4.1 User-Facing Features

1. **Smart Search & Booking** Users can search for and book hotels, flights, and local tours directly within the platform. The search system integrates live data from travel APIs, enabling users to view real-time prices, availability, and location-based recommendations.
2. **AI Travel Agent (LLM Chat)** A conversational AI assistant powered by a Large Language Model (LLM) allows users to describe their travel goals naturally. The agent provides detailed suggestions for destinations, itineraries, accommodations, and activities — adapting dynamically to user feedback and preferences.
3. **Personalized Trip Planner** The AI automatically generates tailored 3–7 day travel plans that include detailed schedules, recommended attractions, and daily activity breakdowns. Users can modify or regenerate itineraries by simply chatting with the AI (“Add one more day in Da Nang”, “Focus more on cultural experiences”, etc.).
4. **User Profile & History** The platform maintains user profiles containing travel history, preferences, and booking records. This information enables the system to continuously improve its recommendations and deliver a more personalized experience over time.
5. **Review & Share** Users can write reviews of hotels, tours, and destinations they have visited. They can also share itineraries with others, promoting community engagement and helping future travelers make informed decisions.

4.2 Backend & Technical Features

- **External API Integration:** Connects to multiple travel data providers (e.g., Agoda, Skyscanner, Amadeus) to retrieve real-time hotel, flight, and activity information. Supports multi-source architecture to reduce dependency on a single API.

- **LLM Integration:** Implements a conversational layer using a Large Language Model (e.g., GPT-4) for entity extraction, context management, and intelligent itinerary generation.
- **Recommendation Engine:** A hybrid system combining content-based and collaborative filtering, enhanced with heuristic rules (distance, budget, rating) to rank and personalize travel suggestions.
- **Database Management:** Stores user profiles, booking histories, and itineraries in scalable and secure databases. Combines relational data (PostgreSQL) with document storage (MongoDB) for flexibility.
- **Authentication & Authorization:** Provides secure user authentication and access control using OAuth 2.0 and JWT, with support for third-party logins (Google, Apple, Facebook).
- **Performance Optimization:** Enhances responsiveness through asynchronous processing, caching, and scalable architecture (e.g., Docker and autoscaling). Includes monitoring and alert systems for performance tracking.
- **Documentation & Deployment:** Includes detailed API documentation, deployment checklists, automated testing pipelines, and compliance measures for privacy and data protection.

5 Technical Challenges and Learning Goals

5.1 Technical Challenges

1. **Integrating the LLM with Real-World Travel Data** The system must process user input and contextual information such as destination, travel dates, and budget so that the AI can understand and respond accurately. → **Solution:** Implement prompt engineering and data pre-processing techniques to extract relevant entities and maintain conversation context.
2. **Developing an Automated Itinerary Recommendation System** Designing a scoring and ranking model to select suitable destinations, activities, and accommodations based on user preferences. → **Solution:** Combine LLM reasoning with heuristic algorithms (e.g., distance, interest matching, popularity score) to generate optimized itineraries.
3. **Synchronizing Data Between APIs and the Backend** Third-party APIs may change endpoints, limit requests, or experience downtime, affecting reliability. → **Solution:** Implement caching strategies, query optimization, and periodic background updates via scheduled cron jobs to ensure data consistency.

5.2 Learning Goals

- Master the **integration of LLM APIs** into a functional web application.

- Strengthen skills in **React/Next.js**, particularly component composition and state management.
- Gain hands-on experience in **combining frontend, backend, and AI systems** into a complete, production-ready solution.
- Learn how to apply **prompt engineering, data modeling, and context handling** for intelligent conversational interfaces.

5.3 Feasibility and Timeline

ID	Phase	Duration	Key Deliverables
1	Research & Design	2 weeks	UX flow, wireframes, and system architecture design
2	Core Development	4 weeks	Implementation of booking module and AI chat integration
3	Testing & Optimization	2 weeks	Unit testing, model refinement, and performance tuning
4	Final Presentation	1 week	Working demo and complete project report

6 Expected Outcome

The final product will be a **AI-powered travel booking website** capable of:

- Searching and booking real hotels and flights.
- Engaging in natural language conversations with users to generate and adjust travel plans.
- Producing detailed and flexible multi-day itineraries tailored to each user's preferences.

The result will be a **fully working prototype** that demonstrates the practical application of **Large Language Models (LLMs)** in the travel and tourism industry — showcasing both innovation and technical feasibility.

7 Future Improvements

- Integration of a **multi-agent architecture** (e.g., AI Planner + AI Concierge) for advanced travel coordination.
- Implementation of **direct booking and payment** capabilities via services such as Stripe or PayPal.
- Incorporation of **machine learning models** to analyze user behavior and continuously improve travel recommendations.
- Expansion to a **mobile app version** for enhanced accessibility and on-the-go trip management.

8 Evaluation Criteria Alignment

Criteria	Description	How the Project Meets It
Clarity & Depth	Project objectives and functionalities are well-defined and thoroughly described.	Clear modular structure and detailed documentation of each feature.
Feasibility	Achievable within the 6–8 week timeline given the current technical skill level.	Includes a structured timeline with defined milestones and deliverables.
Creativity	Innovative combination of AI and travel booking systems.	Utilizes LLMs for personalized travel planning and conversational interaction.

9 Conclusion

This project aims to **redefine the online travel experience** by merging the convenience of modern booking platforms with the intelligence of conversational AI. Users will not only be able to book hotels or flights but also **interact naturally with an AI assistant** to plan entire trips tailored to their needs.

The system will serve as a **proof of concept** demonstrating how **web development and applied AI** can work together to create innovative, user-centric digital products. It highlights the team’s creativity, technical competence, and forward-thinking approach in solving real-world problems through technology.